

narrowed these down to make a comprehensive list. This article delves into some common sensors used in aircraft for various purposes.

Types of sensors

Flow sensors. These are used to monitor the quantities of lubrication oil and liquid coolant fluid in fuel transfer and bleed air systems. Firms like Esterline Corp. and Crane Aerospace & Electronics manufacture liquid sensors and fuel flow sensors. Sensata Technologies Inc. manufactures airflow sensors.

Pressure sensors. Pressure sensors monitor the pressure in hydraulic systems, braking, raising and lowering landing gear, engine oil, oxygen tanks, heating and coolant fluids. Firms like Esterline Corp. and Custom Control Sensors make pressure sensors for aviation.

Temperature sensors. Temperature sensors monitor the conditions of hydraulic oils, fuels, refrigerants and environmental cooling systems. These sensors include bi-metallic temperature gauges, thermometers, Wheatstone bridge indicators, ratiometers and thermocouples.

Omega Engineering Inc. makes high-precision temperature sensors for reliable, easy-to-assemble, extended-life and pre-flight applications. Hydra-Electric makes pressure and temperature sensors based on latest technology.

Altimeters. These measure changes in static air pressure to determine the altitude of the aircraft. For example, MS5803-02BA, based on MEMS technology, is a high-resolution altimeter sensor from TE Connectivity. It includes a high linearity pressure sensor and an ultra low power 24-bit $\Delta\Sigma$ ADC with internal factory-calibrated coefficients.

Airspeed indicators. These calculate true air speed based on pitot tube, static pressure and temperature data.

Position sensors. Position sensors such as linear variable differential transformers (LVDTs) and rotary variable differential transformers (RVDTs) sense the displacement of aircraft components. Companies that make such sensors include Magnetic Sensors



Fig. 3: MS5803-02BA altimeter module (Credit: <https://shop.controleverything.com>)



Fig. 4: A typical attitude heading and reference system (Credit: <http://watson-gyro.com>)



Fig. 5: A typical attitude heading and reference system module

Corp., BEI Sensors and Active Sensors.

Oxygen sensors. Oxygen sensors are at the very heart of the control of inerting systems in Airbus and other civilian aircraft. Different types of oxygen sensors are available from SST Sensing Ltd.

Force and vibration sensors. Such sensors are used on aircraft to measure torque and force in braking and actuation systems as well as in flight controls.

Compasses and magnetometers. These are extremely useful for indi-

cating where the aircraft is headed by measuring Earth's magnetic fields.

Gyroscopes. Gyroscopes are used for direction indication as well as controlling the turning and attitude of aircraft. MEMS gyroscopes are used in modern aircraft. Watson Industries makes different gyroscopes for the aviation industry.

Attitude heading and reference systems. Attitude heading and reference systems (AHRSeS) have replaced most gyroscopes in modern aircraft. Data from MEMS devices, GPSes, magnetometers and accelerometers, and attitude information are received by AHRSeS. Watson Industries also make AHRSeS for aviation. A typical AHRSeS module is shown in Fig. 5.

Tachometers. Tachometers indicate engine rpm. Tachometer probes are used in turbine engines. These sense the changes in magnetic field flux density, as rotating gear wheels move at the same speed as compressor shafts travelling through the probes' magnetic field. Resulting voltage signals are directly proportional to engine speed.

Other aircraft instruments include aneroid barometers, direction indicators, artificial horizons, attitude indicators, laser sensors, sound sensors, IR and RF sensors, among others.

The sensors are either mounted or placed inside the engine to measure various internal and external environmental conditions. These are designed and manufactured with high-reliability and specification standards set by the aviation industry.

Sensor sub-systems

The above-mentioned sensors can be found in the following sub-systems.

Communication. Communications connect the flight deck to the ground, and also the passengers onboard. Very high frequency (VHF) radio band is used for line-of-sight communication such as aircraft-to-aircraft and aircraft-to-air traffic control.

High frequency (HF) radio is used for trans-oceanic flights or satellite communication. Various electronic sensors for communication and navigation systems are available with Bharat Electronics Ltd.

Navigation. Navigation is the determination of position and direction on